

SamuelMATIA

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🎂 16 March 2004 (22 years old)

Education

- Aug 2021 – Sep 2025 **B.Sc., Computer Engineering**, *University of Kinshasa*, Kinshasa
Graduated **top of class (79%)**.
Bachelor Thesis: "Enhancing Sentiment-Driven Recommender Systems With LLM-Based Feature Engineering: A Case Study in Drug Review Analysis" ([IEEE Access](#)), graded **20/20**.
Relevant Coursework: Calculus I-II, Differential Geometry, Linear Algebra and Vector Calculus, Probability and Statistics, Applied Mathematics, Algorithms and Programming, Human-Computer Interaction.
- Completed 2020 **Secondary School, Biochemistry**, *Institut Cardinal Malula*, Kinshasa
Valedictorian (ranked 1st). National Exam: **79%**.

Publications

Journal Articles

- 2025 **Samuel Matia Kangoni**, Obed Tshimanga Tshipata, Pierre Sedi Nzakuna, Vincenzo Paciello, Jean Gilbert Mbula Mboma, Jean Robert Makulo, Kyandoghere Kyamakya. *Enhancing Sentiment-driven Recommender Systems with LLM-Based Feature Engineering: A Case Study in Drug Review Analysis*. **IEEE Access**, 2025. DOI: [10.1109/ACCESS.2025.3590326](https://doi.org/10.1109/ACCESS.2025.3590326).

Conferences

- 2025 **Samuel Matia Kangoni**, Obed Tshimanga Tshipata, Jean-Gilbert Mbula Mboma, Pierre Sedi Nzakuna, Vincenzo Paciello, Vianney Witesyavwirwa Kambale, Kyandoghere Kyamakya. *Structural Preservation in Efficient Pharmacovigilance Summarization: A Causal Graph-Based Evaluation Framework*. **IEEE AFRICON**, 2025. [pre-proceedings draft here](#), official IEEE proceedings pending publication.
- 2025 **Samuel Matia Kangoni**, Mahmoud Hamed, Asha Choudhary, Kyandoghere Kyamakya. *Large Language Models in Healthcare: Opportunities, Challenges, and Pathways Toward Trustworthy Clinical AI*. Accepted at the **AUTSYS-2025: 17th International Conference on Autonomous Systems**. To appear in Springer proceedings.
- 2025 Henri Kasongo Shabani, Kasra Mortazavi, Witesyavwirwa Vianney Kambale, Mahmoud Hamed, **Samuel Matia Kangoni**, Kyandoghere Kyamakya. *A Neuro-Symbolic Edge Stack for Fragile Economies – Binary Cellular Neural Networks and Auto-Mined ASP Rules for Robust Econometrics in the Democratic Republic of Congo*. **WSEAS TRANSACTIONS on BUSINESS and ECONOMICS**. DOI: [10.37394/23207.2026.23.20](https://doi.org/10.37394/23207.2026.23.20)

Book Chapters

- 2025 **Samuel Matia Kangoni**. *Machine Learning in Malware Analysis*. Chapter 7 in *Mastering Malware Development and Analysis: A Comprehensive Guide to Hybrid Malware Analysis and Virtual Sandboxing*. Editors: Bishwajeet Pandey, Deepak Bhaskar Acharya, Divya B. Published by **Apple Academic Press** (Publication: December 2026). ([link](#)) [Publisher page](#) (Hard ISBN: 9781779646224; E-Book ISBN: 978-1-77964-623-1).

Research Experience

- Oct 2025 – Present **Research Mentee – BASIRA (Brain And Signal Analysis Research) Lab – GNNs for Rising Stars**, *Imperial College London*, Supervisor: [Dr. Islem Rekik](#)
Selected for a highly competitive international research mentorship program. Completed the program with **highest distinction** in the final examination. Conducted advanced work on Graph Neural Networks (GNNs), with a focus on architecture design, explainability, and reproducible machine learning. Collaborated within an international cohort on a publication-oriented project. [BASIRA Lab web page](#)

Sep 2025 – **Researcher – BASTION Lab**, *University of Kinshasa*, Supervisor: [Dr. Kyandoghene Kyamakya](#)
Present Researching neuro-symbolic AI for robust and explainable systems in challenging contexts. Developing hybrid architectures that integrate neural networks (Binary Cellular Neural Networks, BD-CeNN Autoencoders) with symbolic reasoning (Answer Set Programming) for applications in econometric modeling and medical imaging.

Selective Programs

2026 **A2SV (Africa to Silicon Valley) – G7 Remote Education Program**, *Remote*
Selected among [166 out of 2,264 applicants](#) (~ 7% acceptance rate). Elite training program in algorithmic problem solving and data structures, aimed at preparing high-performing African students for top-tier software engineering and research roles. Engaged in intensive daily problem-solving ([LeetCode](#)), advanced data structures, and weekly Codeforces contests under competitive programming constraints.

Professional Experience

Aug 2025 – **Freelance AI Trainer**, *EAZYTraining / EASYCONSULTING*, Remote (France)
Present Designed and delivered online AI/DS courses with hands-on projects and deployment workflows; developed tailored training content on ML, DL, and Generative AI.

Jul 2024 – Mar 2025 **AI Engineer**, *CongoExcelTechnologies*, DR Congo,
Built end-to-end RAG solutions (vector DBs, web scraping) for custom knowledge systems; integrated third-party APIs/databases; benchmarked LLMs, embeddings, and fine-tuning strategies.

Sep 2024 – Nov 2024 **AI Engineer Intern**, *PathOr Platforms*, Remote (India)
Developed task-specific LLM agents (shopping assistance, data analysis, automation).

Dec 2023 – Mar 2024 **AI Consultant**, *IA Flex Start*, Remote (France)
Integrated LLMs into production apps via custom APIs; conducted end-to-end data preprocessing.

Leadership and Community Engagement

Sep 2024 – Oct 2025 **Country Ambassador (DRC)**, *Zindi Africa*
Built and led a 200+ member AI community; organized hackathons and technical workshops to promote DataScience and machine learning.

Jul 2025 – Jul 2026 **Project Coordinator & Informatics Lab Manager**, *Polytechnic Faculty, University of Kinshasa*
Managed lab/server infrastructure; mentored student projects in AI/CS.

Jul 2023 – Jul 2024 **Co-lead**, *Google Developer Students Club – University of Kinshasa*
Coordinated technical events/workshops; mentored students in software development and AI.

Awards and Distinctions

2026 **TReND in Africa Computational Neuroscience & Machine Learning Summer School** – Awarded a full scholarship after selection among [20 out of 1,060 applicants](#) (~ 1.9% acceptance rate) for a highly competitive international research training program in computational neuroscience and machine learning. Offer later declined due to overlapping international research admission procedures ([Invitation letter](#)).

2024 **Hackathon TechPoly 2024 – First Prize**. *Mat.IA*: multilingual tutor web app combining computer vision, LLM dialogue, and TTS.

2024 **Huawei Seeds for the Future** – Selected as one of 10 students to represent DR Congo in the ICT flagship program. ([Certificate of Completion](#))

2023 **Orange Summer Challenge 2023 – Second Prize**. Built the “Utulivu” autism screening system (robot + wearable + app); deployed object detection on Raspberry Pi; created a hybrid recommender system.

Skills

Concepts	Machine Learning, Deep Learning, NLP, Recommender Systems, Graph Neural Networks, Computer Vision, Reinforcement Learning, Web Development
Frameworks	PyTorch, TensorFlow, NumPy, LangChain, LangGraph, HuggingFace, scikit-learn, OpenCV, Pandas, Matplotlib, Streamlit, Django
Programming	Python, JavaScript, MATLAB, Git, Linux, Docker, HTML/CSS
Research	Problem Solving, Experimentation, $\LaTeX$
Cloud	AWS, GCP

Languages

French, Lingala, Kikongo – Native proficiency

English – Professional working proficiency